

Analysis of Scientific Data



Lecture 7

Comparing Means

The P -value

The p -value is a transformation of the data:

$$\text{Data} \longrightarrow \bar{X} \longrightarrow T \longrightarrow P$$

Thus the p -value is a random variable in the same way that $\bar{X}$ is.

Similarly, a confidence interval procedure generates random intervals.

Poll Question

Suppose a substance (perhaps water?) has no effect on pulse rate ($\mu = 0$) so H_0 is in fact true.

What is the probability that the p -value will be less than 5%?

1. 5%

2. 10%

3. 50%

4. Impossible to tell

Poll Question

If H_0 is true, what is the shape of the p -value distribution?

1. Uniform
2. Normal
3. Skewed to the left
4. Skewed to the right

Type I Error

If we reject the null hypothesis when in fact it is true, we have made a *Type I error*.

The probability of making a Type I error is precisely the significance level α that we choose for making decisions.

Type II Error

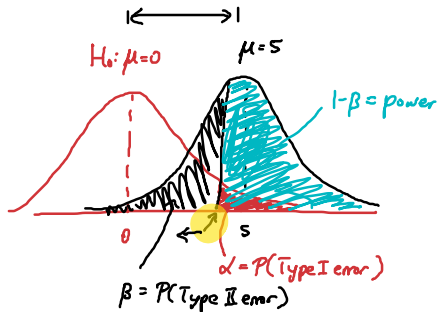
If we do not reject the null hypothesis when in fact it is false, we have made a *Type II error*.

If β is the probability of making a Type II error then the *power* is $1 - \beta$.

Poll Question

Which of the following would **not** increase the power of a study?

1. Decrease sample size
2. Increase effect size
3. Decrease residual variability
4. Increase α



Summary

	Decision	
	Not reject	Reject
H_0 is true	Correct $(1 - \alpha)$	Type I Error (α)
H_0 is false	Type II Error (β)	Correct <i>power</i> $(1 - \beta)$

Example – Tuatara Mass (Final Exam 2020)

A tuatara is a reptile that is endemic to New Zealand and the sole surviving species of its order that otherwise disappeared 100 million years ago.



Since they have evolved in isolation, tuatara are of interest in the study of adaptation. One study investigated two separated populations of tuataras, one at Location A and one at Location B, and looked for any differences between these populations.

Example – Tuatara Mass (Final Exam 2020)

Is there any evidence of a difference in mean mass between tuatara populations at Locations A and B?

Suppose μ_A is the mean mass of tuataras at location A and μ_B is the mean mass of tuataras at location B. The researchers would like to test

1. $H_0 : \mu_A = \mu_B$ vs $H_1 : \mu_A \neq \mu_B$

2. $H_0 : \mu_A = \mu_B$ vs $H_1 : \mu_A < \mu_B$

3. $H_0 : \mu_A = \mu_B$ vs $H_1 : \mu_A > \mu_B$

~~4. $H_0 : \mu_A < \mu_B$ vs $H_1 : \mu_A = \mu_B$~~

Example – Tuatara Mass (Final Exam 2020)

The 9 tuataras measured at location A had a mean mass of 617.7 g with a standard deviation of 38.67 g, while the 14 tuataras measured at location B had a mean mass of 585.8 g with a standard deviation of 23.39 g.

$$H_0: \mu_A = \mu_B \quad \text{vs} \quad H_1: \mu_A \neq \mu_B$$

$$E(\bar{X}) = \mu$$

$$sd(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

$$\text{Observed } n_A = 9, \quad \bar{x}_A = 617.7, \quad s_A = 38.67$$

$$n_B = 14, \quad \bar{x}_B = 585.8, \quad s_B = 23.39$$

$$\text{The difference in mean } \bar{x}_A - \bar{x}_B = 31.9$$

$$\begin{aligned} \text{(one-sided)} \quad p\text{-value} &= P(\bar{X}_A - \bar{X}_B \geq 31.9 \mid H_0 \text{ true}) \\ &= P\left(\frac{(\bar{X}_A - \bar{X}_B) - E[\bar{X}_A - \bar{X}_B]}{sd(\bar{X}_A - \bar{X}_B)} \geq \frac{31.9 - E(\bar{X}_A - \bar{X}_B)}{sd(\bar{X}_A - \bar{X}_B)} \mid H_0 \text{ true}\right) \end{aligned}$$

$$E(\bar{X}_A - \bar{X}_B) = E(\bar{X}_A) - E(\bar{X}_B) = \mu_A - \mu_B$$

$$\text{Assuming independence, } Var(\bar{X}_A - \bar{X}_B) = Var(\bar{X}_A) + Var(\bar{X}_B) = \frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}$$

Example – Tuatara Mass (Final Exam 2020)

$$sd(\bar{X}_A - \bar{X}_B) = \sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}}$$

$$se(\bar{X}_A - \bar{X}_B) = \sqrt{\frac{S_A^2}{n_A} + \frac{S_B^2}{n_B}}$$

$$= P\left(\frac{(\bar{X}_A - \bar{X}_B) - (\mu_A - \mu_B)}{se(\bar{X}_A - \bar{X}_B)} \geq \frac{31.9 - (\mu_A - \mu_B)}{se(\bar{X}_A - \bar{X}_B)} \mid H_0 \text{ true} \right)$$

$H_0: \mu_A = \mu_B = 0$

$$= P\left(T \geq \frac{31.9}{\sqrt{\frac{S_A^2}{n_A} + \frac{S_B^2}{n_B}}} \right)$$

$$\begin{aligned} df &= \min(n_A - 1, n_B - 1) \\ &= \min(8, 13) \\ &= 8 \end{aligned}$$

$$= P\left(T \geq \frac{31.9}{\sqrt{\frac{38.67^2}{9} + \frac{23.39^2}{14}}} \right)$$

$$= P\left(T \geq \frac{31.9}{14.33} \right)$$

$$p\text{-value} = 1 - pt(2.227, df=8)$$

$$= P(T_8 \geq 2.227)$$

$$= 0.0283$$

Two-sample t -tests

$$\begin{aligned}\text{two-sided } p\text{-value} &= 0.0283 \times 2 \\ &= 0.0566\end{aligned}$$

Assumptions:

Sample 1: $X_{1,1}, X_{1,2}, \dots, X_{1,n_1} \stackrel{iid}{\sim} \text{Normal}(\mu_1, \sigma_1)$

Sample 2: $X_{2,1}, X_{2,2}, \dots, X_{2,n_2} \stackrel{iid}{\sim} \text{Normal}(\mu_2, \sigma_2)$

- ① Normality
- ② Independent

Test $H_0 : \mu_1 = \mu_2$.

We consider three situations:

- **Welch t -test**: Assume $\sigma_1 \neq \sigma_2$, and the two samples are independent.
- **Pooled t -test**: Assume $\sigma_1 = \sigma_2$, and the two samples are independent.
- **Paired t -test**: Assume $n_1 = n_2$, and the samples have paired structure.

Welch t -test

- Test statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

- When H_0 is true,

$$T \stackrel{\text{approx.}}{\sim} t_{df},$$

where

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}} \approx \min(n_1 - 1, n_2 - 1).$$

- p -value =

- ▶ $2P(t_{df} > |t|)$ if $H_1 : \mu_1 \neq \mu_2$
- ▶ $P(t_{df} > t)$ if $H_1 : \mu_1 > \mu_2$
- ▶ $P(t_{df} < t)$ if $H_1 : \mu_1 < \mu_2$

$$CI = \text{estimate} \pm t_{df}^* se(\text{estimate})$$

$$= (\bar{x}_1, \bar{x}_2) \pm t_e^* 14.33$$

$$qt(0.975, df=8) = 2.306$$

$$\min(n_1 - 1, n_2 - 1) = 31.9$$

$$\uparrow \text{ use this in STAT201 } \pm 2.306 \times 14.33$$

$$= 31.9 \pm 33.05$$

- Test statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

where the pooled sample variance s_p^2 is

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

- When H_0 is true,

$$T \sim t_{n_1 + n_2 - 2}$$

- p -value =

- ▶ $2P(t_{df} > |t|)$ if $H_1 : \mu_1 \neq \mu_2$
- ▶ $P(t_{df} > t)$ if $H_1 : \mu_1 > \mu_2$
- ▶ $P(t_{df} < t)$ if $H_1 : \mu_1 < \mu_2$

Paired t -test

- The data:

Sample 1: $X_{1,1}, X_{1,2}, \dots, X_{1,n}$

Sample 2: $X_{2,1}, X_{2,2}, \dots, X_{2,n}$

- Take the difference $\underline{D_i} = X_{1,i} - X_{2,i}$ for $i = 1, 2, \dots, n$.
Assume

$$\boxed{D_1, D_2, \dots, D_n} \stackrel{iid}{\sim} \text{Normal}(\mu_1 - \mu_2, \sigma)$$

- Proceed as a one-sample t -test.

Piglets and Pertussis

Pertussis continues to be a significant cause of morbidity and mortality in infants and young children worldwide. Researchers used a piglet model of human immunity to determine whether maternal immunisation could help prevent this disease in infants. Twelve pregnant sows were vaccinated four weeks prior to farrowing and boosted two weeks later, with another twelve pregnant sows used as a control. In each group of twelve there were three different breeds of pig, four Landraces, four Large Blacks and four Hampshires. A blood sample was taken from the first-born piglet from each sow prior to suckling. Serum concentrations (mg/dL) of pertussis-specific IgG antibodies were measured by ELISA. The sex and weights (kg) of the piglets were also recorded.

Analysis of Variance (ANOVA)

3 questions we can ask:

- ① Is there a difference in mean serum concentrations (mg/dL) of pertussis-specific IgG antibodies between the three breeds?
- ② Does the vaccine have an effect on serum concentrations (mg/dL) of pertussis-specific IgG antibodies?
- ③ Is the vaccine effect the same for all breeds?

We focus on ① today

Is there a difference in mean serum concentrations (mg/dL) of pertussis-specific IgG antibodies between the three breeds?

Let μ_H = mean antibody concentration for Hampshire
 μ_L = " " " " " Landrare
 μ_B = " " " " " Large Black

Test $H_0: \mu_H = \mu_L = \mu_B$ vs H_1 : not all means equal

Total Variability

$$S^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$= \frac{6504450}{28} \leftarrow \begin{array}{l} \text{total sum of square (SST)} \\ \text{total df (DFT)} \end{array}$$

Residual Variability

Assuming homogeneous variance, pool together residuals from each group

sum of square residual (SSR) = 5503632

residual DF DFR = $n - 3 = 21$

ANOVA table

Source	DF	SS	MS	F	P
Breed	2	1000819	500409.5	1.909	0.173
Residual	21	5503632	262077.7		
Total	23	6504451			

mean square (i.e. variance) $F = \frac{MSG}{MSR}$

$P(F_{2,21} > 1.909)$

In R: $1 - pf(1.909, 2, 21)$

Poll Question

What are the **total** degrees of freedom for the IgG concentrations?

1. 18

2. 19

3. 20

4. 21

5. 22

6. 23

7. 24

12 vaccine

12 control

$n=24$

Poll Question

What are the **residual** degrees of freedom for the one-way ANOVA of IgG concentration by breed?

1. 18

2. 19

3. 20

4. 21

5. 22

6. 23

7. 24

df for each group is $8 - 1 = 7$

There are 3 groups (3 breeds).

So total is $7 \times 3 = 21$.

ANOVA Summary

Assume:

- Sample 1: $X_{1,1}, X_{1,2}, \dots, X_{1,n_1} \stackrel{iid}{\sim} \text{Normal}(\mu_1, \sigma)$
- Sample 2: $X_{2,1}, X_{2,2}, \dots, X_{2,n_2} \stackrel{iid}{\sim} \text{Normal}(\mu_2, \sigma)$
- $\vdots$
- Sample k : $X_{k,1}, X_{k,2}, \dots, X_{k,n_k} \stackrel{iid}{\sim} \text{Normal}(\mu_k, \sigma)$

Test $H_0 : \mu_1 = \dots = \mu_k$ vs $H_1: H_0$ not true.

Source	DF	SS	MS	F	p -value
Groups	$DFG = k - 1$	SSG	$MSG = \frac{SSG}{DFG}$	$f = \frac{MSG}{MSR}$	$P(F_{DFG, DFR} > f)$
Residuals	$DFR = n - k$	SSR	$MSR = \frac{SSR}{DFR}$		
Total	$n - 1$	SST			